ANALYSIS OF SUPPLY

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Supply and Stock

- **Stock** is the amount of goods currently available for sale.
- Stock: regarded as inventories.
- **Supply**: is the amount of goods that sellers offer to sell at various prices, ceteris paribus.
- Rational seller: maximises profits
- Supply is a flow of goods at various prices,
- Can be shown as a schedule or in a diagram.

The Supply Curve

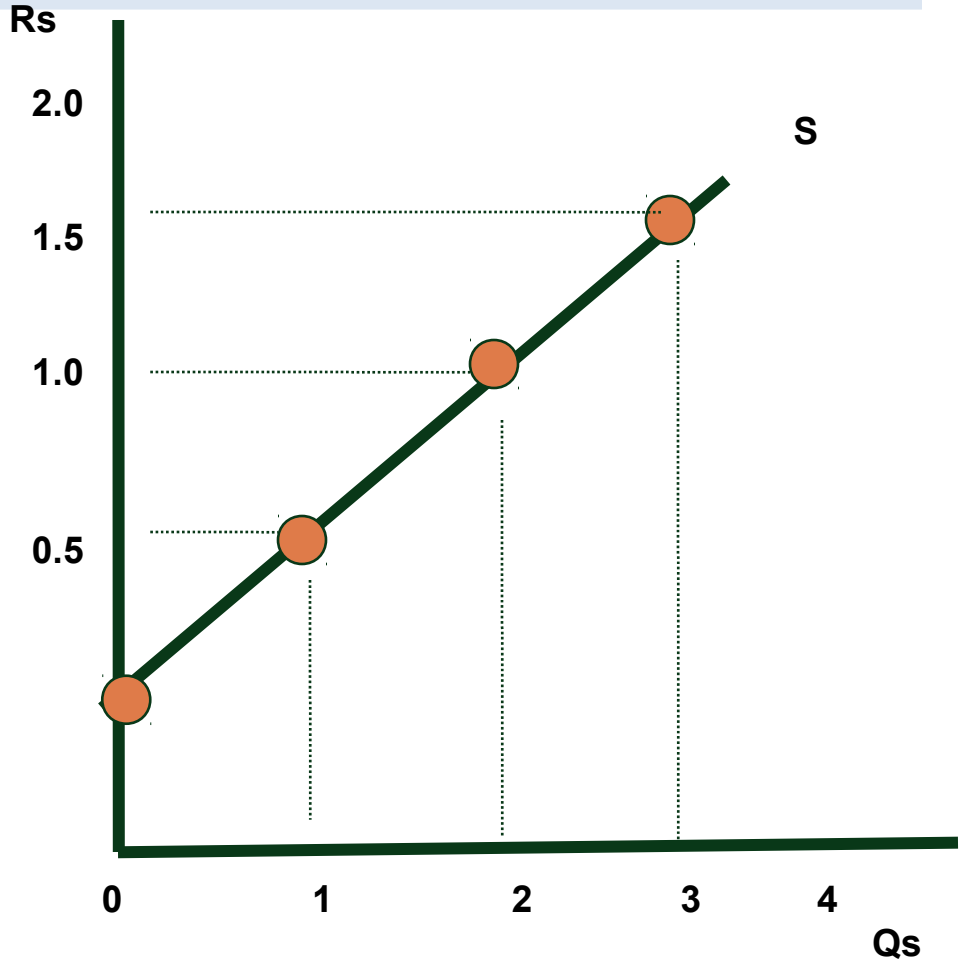
- **The Law of Supply:** Other things remaining constant, quantity supplied of a commodity is positively related to its price.

$$S_q = f(P)$$

- Higher the price, higher the quantity supplied.
- The Supply curve slopes upwards to the right.
- Higher price means higher profit margin, given costs.
- So sellers increase their supply, when price is higher.

Supply Curve

Price (Rs)	Q supplied (kgs)
0.5	1
1.0	2
1.5	3
2.0	4
2.5	5
3.0	6
3.5	7



Factors affecting Supply

1. Price of the commodity – *more is supplied at higher prices, less at lower prices.*
2. Prices of related goods –

   Substitutes - If price of tea falls, then demand for tea increases, but demand for coffee falls. So supply of coffee falls.

   Complements in production -- good that is produced with other good – e.g. price of cars decrease, demand and so supply of tires increases.

Factors affecting Supply

3. Seller's expectations, Expect input prices to fall in future: *increase supply today*

   Expect price of good to rise in future: *decrease supply today*
4. Number of sellers is more, market supply increases
5. Technology – *new technology, new products, so S of old products falls.*
6. Time Period – *more time, higher supply*
7. Prices of inputs – *or cost of production increases e.g. wages, affects supply decreases.*

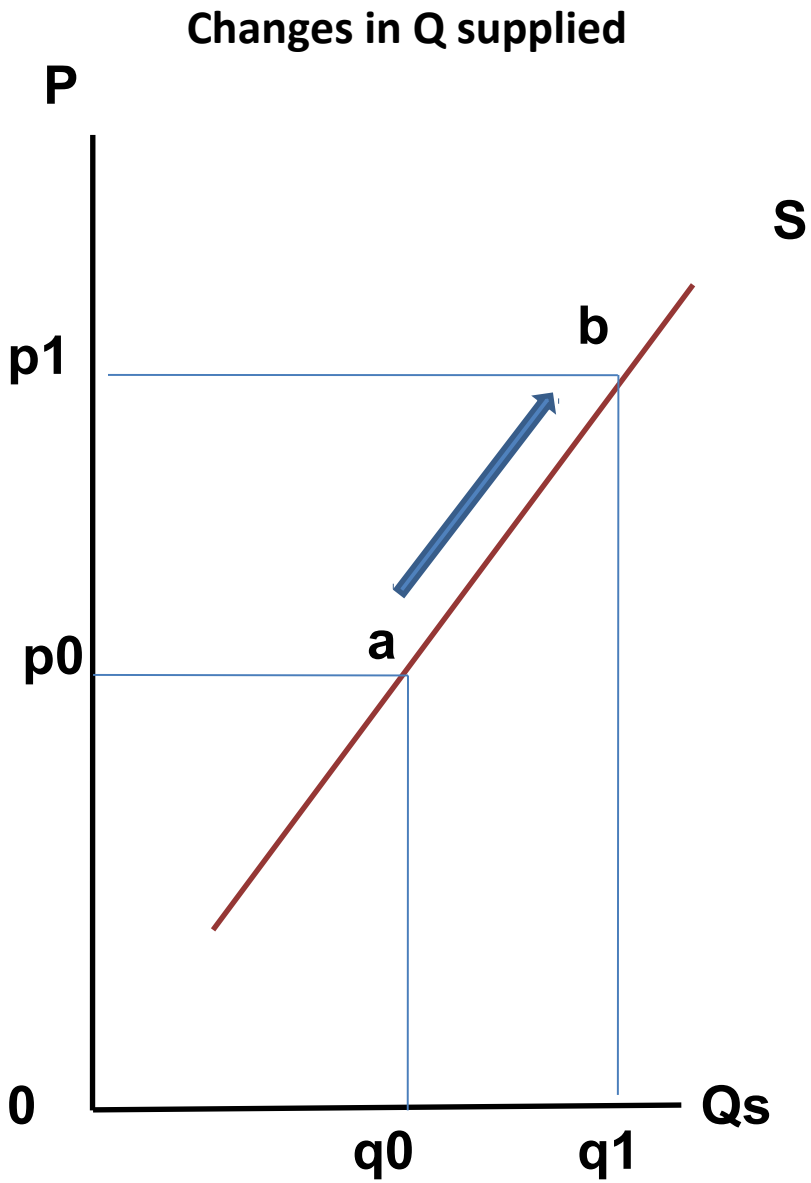
Changes is Quantity Supplied vs. Change in Supply

1. **Changes in Quantity supplied**: also called movements along a given supply curve.

   Affected by changes in the price of the commodity only, ceteris paribus.

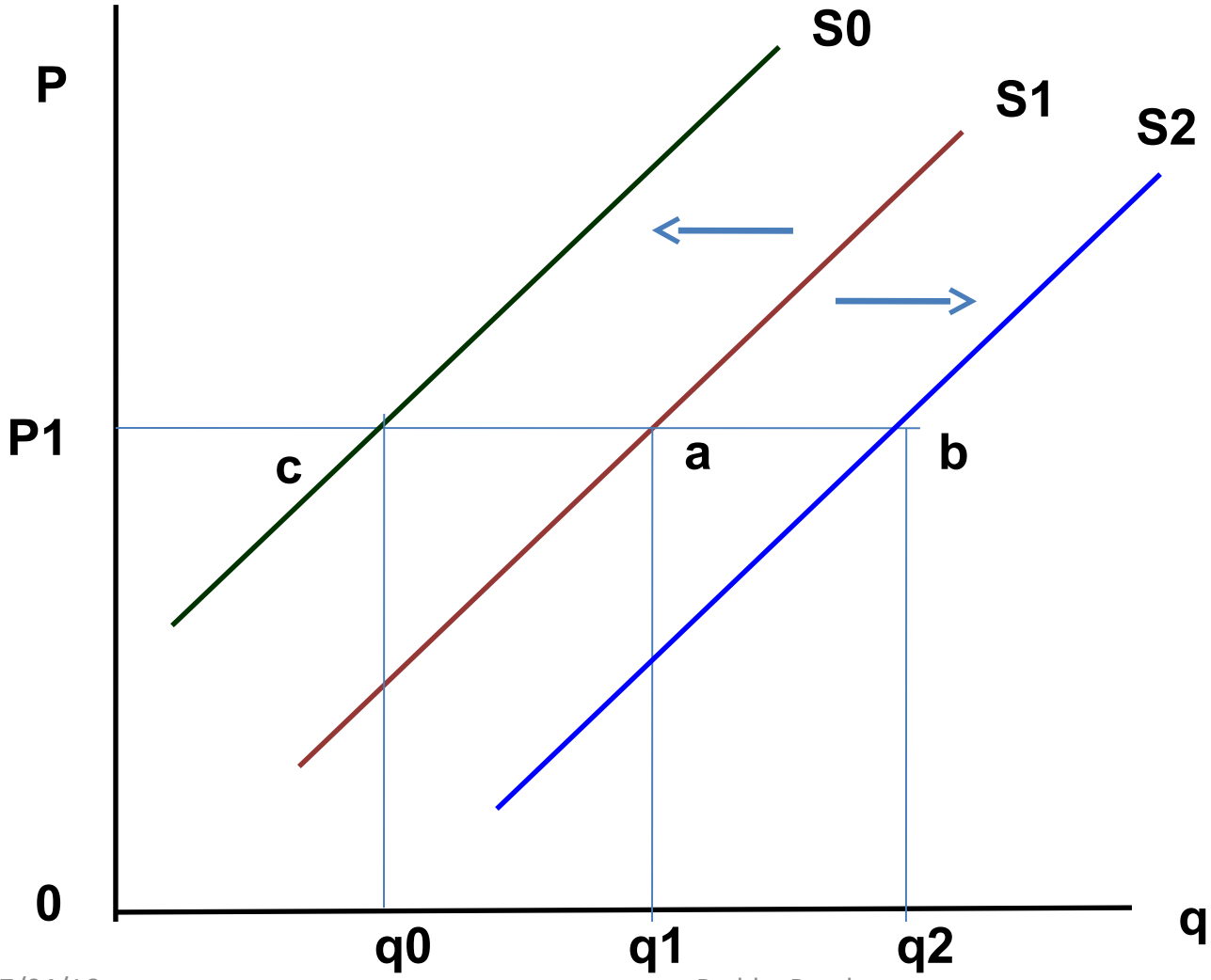
2. **Change in Supply**: or shift in the Supply curve.

   Changes in other factors affecting supply, causes entire supply curve to shift upwards or downwards.



Change in Q supplied:
On a given Supply curve, any increase or decrease in price, will cause quantity supplied to increase or to decrease.

Shifts in Supply curve – Increase and Decrease in Supply



If ceteris paribus conditions do not exist, then the total supply curve may shift. E.g. change in technology, or taxes, etc.
It can shift to the right (increase in supply) or to the left (decrease in supply).

Market Supply

- The sum total of all supply schedules of different sellers at each price, gives the Market Supply.
- Market Supply is the horizontal summation of individual sellers, at different prices.

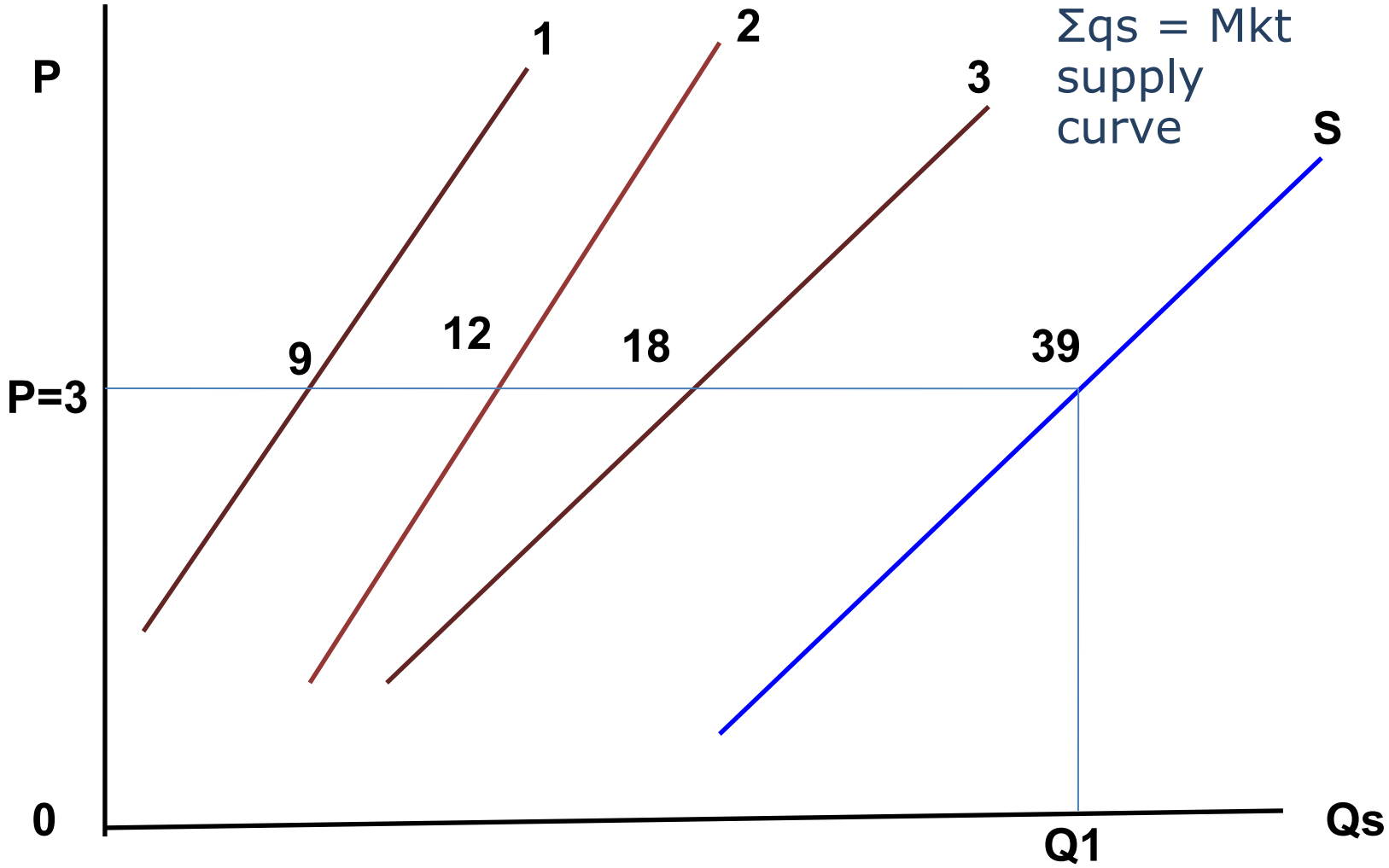
$$\mathbf{S} = \Sigma S$$

Market price is determined by Market Supply and Market Demand curves.

Market Supply

Px	q1	q2	q3	Market S = Σqs
1	5	8	12	25
2	7	10	15	32
3	9	12	18	39
4	11	14	21	46
5	13	16	24	53

Market Supply - Diagram



Elasticity of Supply

Elasticity of Supply

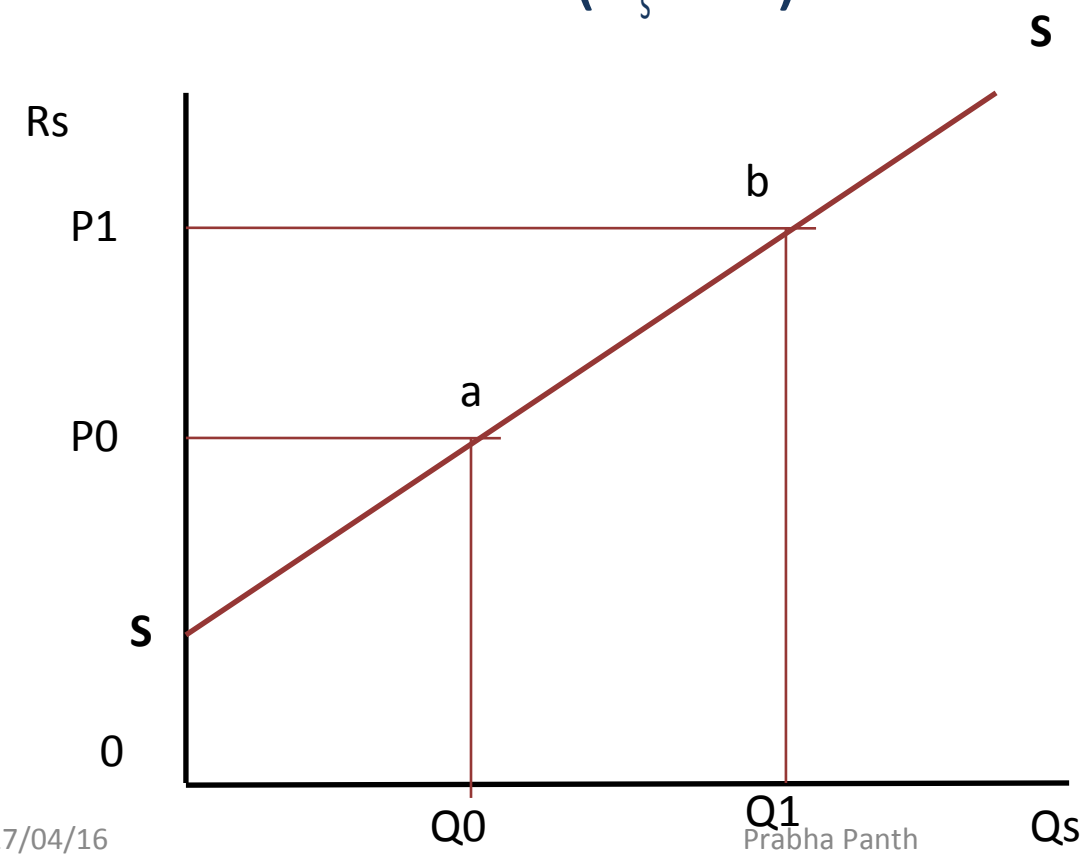
- Elasticity of Supply is the responsiveness of quantity supplied to a change in the price, ceteris paribus.
- It is given by the formula:

$$+ e_s = \frac{\Delta Qs.}{\Delta P} \frac{P}{Qs}$$

Usually the elasticity of supply is a positive number, showing a positive response of quantity supplied to a change in the price.

Types of S-elasticities

1. <u>Elastic supply:</u> Rate of increase in Qs > rate of increase in P. ($e_s > 1$)



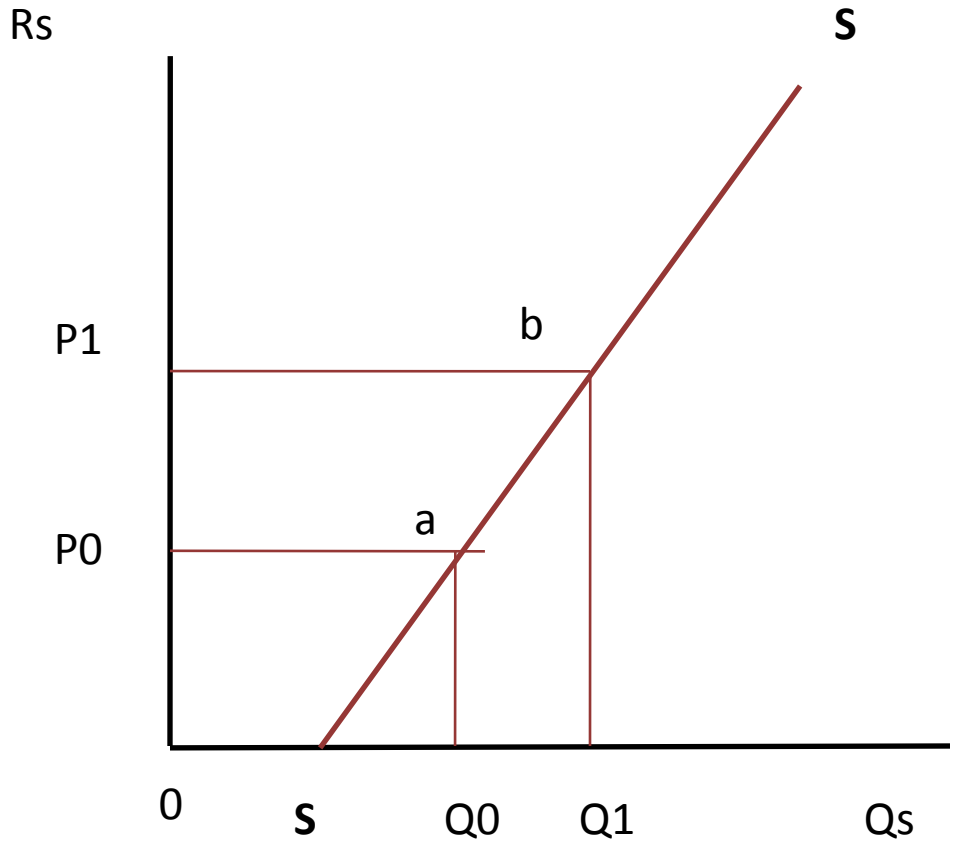
Supply responds at a rate greater than rate of change in P. Goods whose supply can easily increase, or are available in stock.

2. Inelastic supply:

Qs responds at a rate lower than change in price.

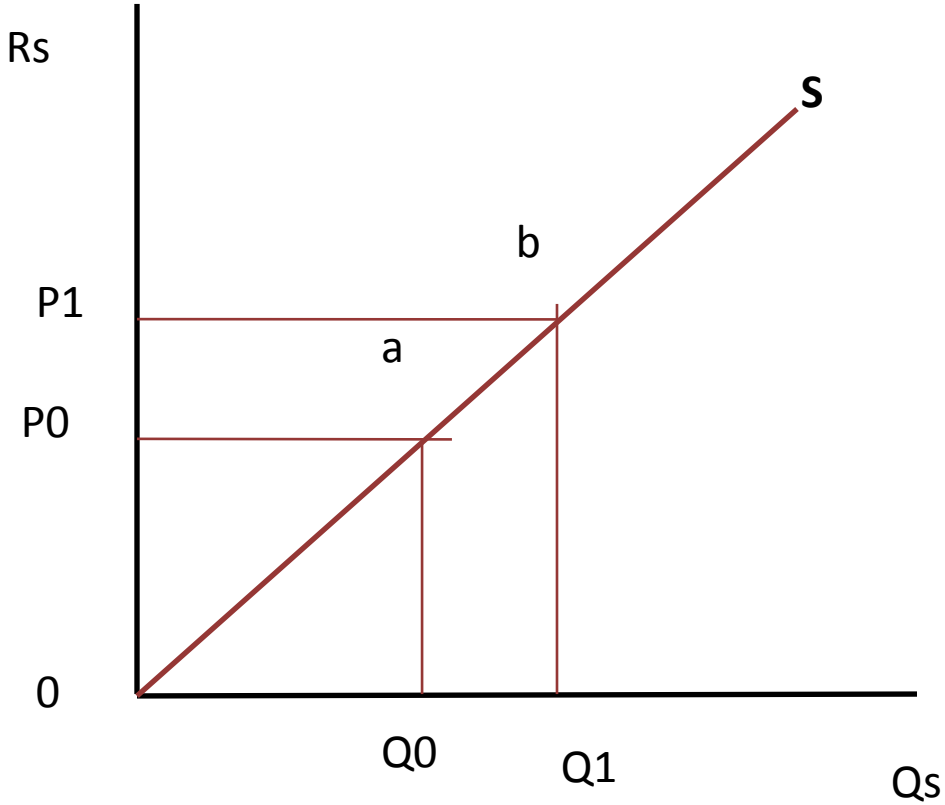
($e_s < 1$)

Goods needing more time for production, or low stocks have inelastic supply.

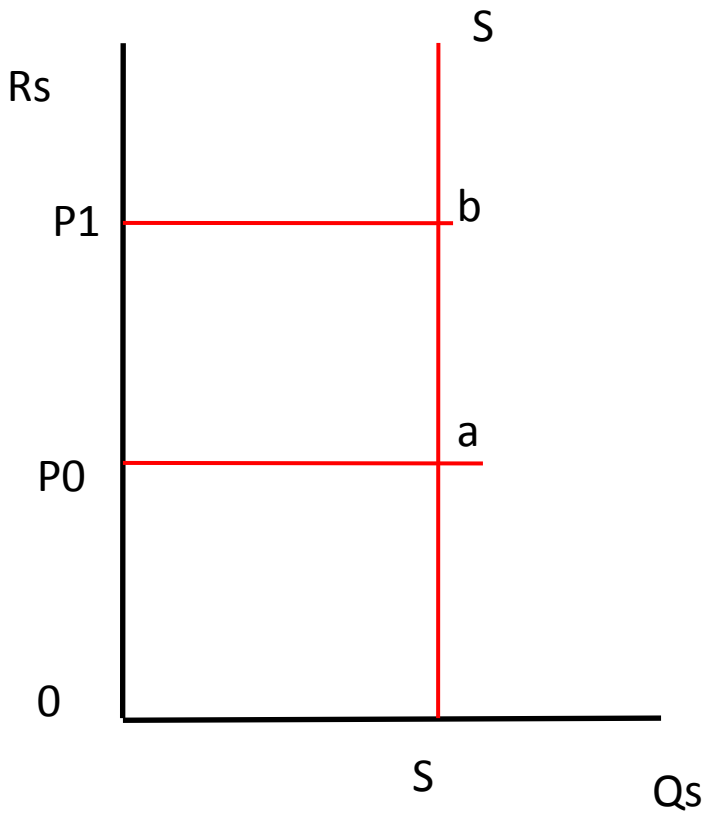


3. <u>Unitary elasticity</u>: $e_s = 1$, and quantity supplied responds at the same rate as change in price.

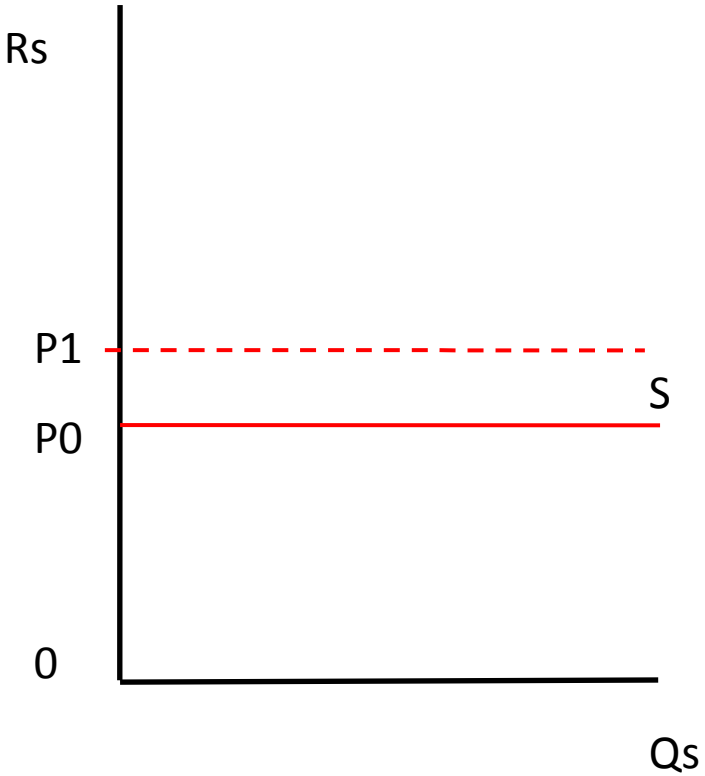
$\Delta Q/Q = \Delta P/P$



4. Perfectly inelastic: There is no response of Qs to ΔP. ($e_s = 0$)



5. Perfectly elastic: or infinite elasticity. Supply responds infinitely to ΔP. ($e_s = \propto$)



Questions

I. Short answers:

a) What is the difference between Stock and Supply?

b) State and explain the Law of Supply.

c) What are the factors that affect supply?

d) How is elasticity of supply measured? Give the formula and a diagram to illustrate your answer.

Questions

II. Essay Questions:

1. What is the difference between ‘changes in Qs’ and ‘increase and decrease in S’? Explain with the help of diagrams.
2. What is Market supply? How is it derived?
3. What is elasticity of supply? Illustrate the different types of price elasticities with the help of diagrams.